

Interview Lens

An AI Interview Platform for Mastering Behavioral Interviews

Course Team Project Deliverable #2

SE / CS / CE 3354 Software Engineering, Summer 2026

Team 1

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1. Project Title

Interview Lens: An AI Interview Platform for Mastering Behavioral Interviews.

Interview Lens is a web application that lets college students practice realistic behavioral interviews with an AI interviewer, receive an immediate scored report on how they answered, and track whether they are actually improving over time. The system is built on a React frontend, an Express backend, a PostgreSQL database, and an AI engine that both generates questions and evaluates responses.

2. Delegation of Tasks

Work on Deliverable #2 was divided so that every member owned at least one graded section from start to finish in addition to their engineering area. Every member contributed and every member presents.

Team Member	Deliverable #2 Responsibilities
Ayro S. Escobar	Section 4 in full: project scheduling, Function Point cost estimation, and the hardware, software and personnel cost breakdowns. Maintained the database schema and endpoint logic and owned the class diagram content.
Ben Andre Quidlat	Report assembly and formatting. Sections 1, 2, 3, 7 and 8. Backend integration and the CI/CD pipeline that enforces the coverage gate in Section 5. Owned the GitHub repository structure and the GitHub and eLearning submissions.
Hadeed Sarfaraz	AI pipeline design and prompt logic for behavioral scenarios. Section 6 in full, including the source verification described there. Sequence diagrams and the process model rationale in Section 3.
Shaun Kenan Moukam	Frontend UI/UX components, routing and public deployment. Section 9, the presentation slide deck, and the Figma prototype. Use case diagram and the MVC architecture write-up.
Joshua Asteway	Section 5 in full: the test plan, the unit under test, the automated Jest run, and the defined test cases including the failing case. User acceptance testing on every increment and verification of the non-functional targets.

Note on team size: the original project proposal listed four members. Joshua Asteway joined the team afterwards and took ownership of quality assurance. All Deliverable #2 material reflects the current five member team.

3. Deliverable #1 Content

This section reproduces the full content of Course Team Project Deliverable #1 so that Deliverable #2 stands on its own.

3.1 Assumptions

We make the following assumptions. A3, A4 and A6 are invented regulatory rules, stated here per the deliverable instructions so that every non-functional requirement type can be covered.

- A1. The platform initially targets college students in the United States and supports English language interviews only.
- A2. Because our users are students and the platform may be adopted by universities, interview recordings, transcripts and feedback are treated as educational records and handled in a FERPA aligned manner.
- A3. We assume a U.S. AI Interview Transparency Act of 2025 requiring any platform that gives AI generated hiring or interview feedback to disclose that the feedback is AI generated and to let users flag feedback for human review.
- A4. We assume the same Act requires permanent deletion of all of a user's data within 30 days of a deletion request.
- A5. University single sign on is not required for the first release, so email and password authentication is sufficient.
- A6. We assume an audit retention rule requiring account and data access events to be logged and retained for 12 months.

3.2 Software Process Model

We are using the incremental development process model described by Sommerville [1]. We chose it for four reasons. First, the summer timeline is short, so a usable core system has to ship early. Second, requirements for an AI product are volatile, because how the AI interviewer should behave cannot be fully predicted until people actually use it. Third, each increment produces something working that we can put in front of the instructor and TA, and that feedback shapes the next increment. Fourth, it fits our tooling, since the CI/CD pipeline validates every increment automatically.

We rejected the waterfall model because our requirements are still evolving. We rejected pure integration and configuration because the core of the product, the AI interview loop and the scoring rubric, has to be custom built.

3.3 Functional Requirements

ID	Functional Requirement
FR-1	The system shall allow users to register an account with name, email and password, and to log in and log out. Duplicate emails shall be rejected.
FR-2	The system shall allow a logged in student to start a mock behavioral interview session by selecting a target job role and a difficulty level.
FR-3	The system shall use the AI engine to generate at least five behavioral interview questions per session, including adaptive follow up questions based on the student's previous answer.
FR-4	The system shall accept student answers as typed text or recorded audio, and shall transcribe audio answers to text before analysis.
FR-5	Upon session completion, the system shall generate a feedback report that scores each answer on STAR structure, clarity and confidence, and gives an overall session score from 0 to 100 with written improvement suggestions.
FR-6	The system shall store every completed session and its report, and shall display the student's history and score trends over time on a progress dashboard.
FR-7	The system shall allow an administrator to add, edit and remove questions and question categories in the question bank.

3.4 Non-Functional Requirements

ID	Type	Non-Functional Requirement
NFR-1	Usability	A first time user shall be able to create an account and start their first mock interview in under 3 minutes without training or documentation. Contextual help shall be available on every screen.
NFR-2	Performance	The system shall deliver the next interview question within 2 seconds and the full end of session feedback report within 15 seconds, for 95% of requests under normal load.
NFR-3	Space	The initial web bundle shall not exceed 5 MB, and a completed session, transcript plus report, shall consume no more than 2 MB of database storage on average.
NFR-4	Dependability	The platform shall maintain at least 99.5% monthly availability, and every answer shall be autosaved as it is submitted so that a crash or disconnect loses at most the answer currently in progress.
NFR-5	Security	All traffic shall use TLS 1.2 or higher. Passwords shall be stored only as bcrypt hashes. Session tokens shall expire within 24 hours. Role based access control shall prevent students from accessing admin functions or other students' data.
NFR-6	Environmental	The system shall run in the two most recent versions of Chrome, Firefox, Safari and Edge on desktop and mobile without local installation, and shall be deployable on commodity Linux cloud instances.
NFR-7	Operational	The system shall support at least 500 concurrent interview sessions during peak recruiting season, and administrators shall be able to update the question bank without taking the platform offline.
NFR-8	Development	All code shall be written in TypeScript, held in the team GitHub repository, merged only through reviewed pull requests, and covered by unit tests at a minimum of 70% line coverage enforced by the CI/CD pipeline.

ID	Type	Non-Functional Requirement
NFR-9	Regulatory	Per assumption A2, interview transcripts, recordings and feedback shall be handled as FERPA aligned educational records: private by default and never shared with third parties without the student's explicit consent [10].
NFR-10	Ethical	The platform shall clearly label all feedback as AI generated, shall not sell user data, shall not use student responses to train AI models without explicit opt in consent, and shall review AI prompts for bias against protected groups.
NFR-11	Accounting	Per assumption A6, the system shall keep an audit log of all account events and data access events and retain these logs for 12 months.
NFR-12	Safety and Security (Legislative)	Per assumptions A3 and A4, the system shall comply with the assumed AI Interview Transparency Act of 2025 and shall permanently delete all of a user's data within 30 days of a deletion request.

3.5 Use Case Diagram

Interview Lens has two human actors, Student and Administrator, and one system actor, the AI Engine. We modelled the AI Engine as an actor rather than an internal component because it is an external service we call rather than code we own. That decision carries through to Section 4, where it is counted as an External Interface File rather than an Internal Logical File. Register and Log In is shared by both human actors rather than duplicated, which keeps one authentication path and one place to enforce the role based access control required by NFR-5.

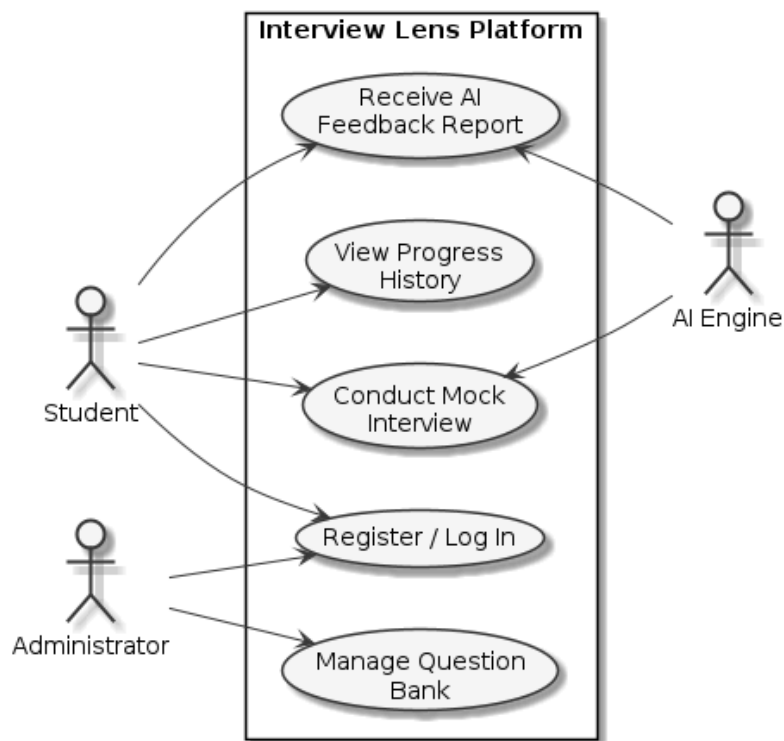


Figure 1. Use case diagram for Interview Lens

3.6 Sequence Diagrams

Five interactions were modelled. Each follows the MVC boundary, running from a UI object through a controller to the AI engine or the database.

3.6.1 Register and Log In. Covers account creation and authentication, with alt fragments for a duplicate email and for invalid credentials.

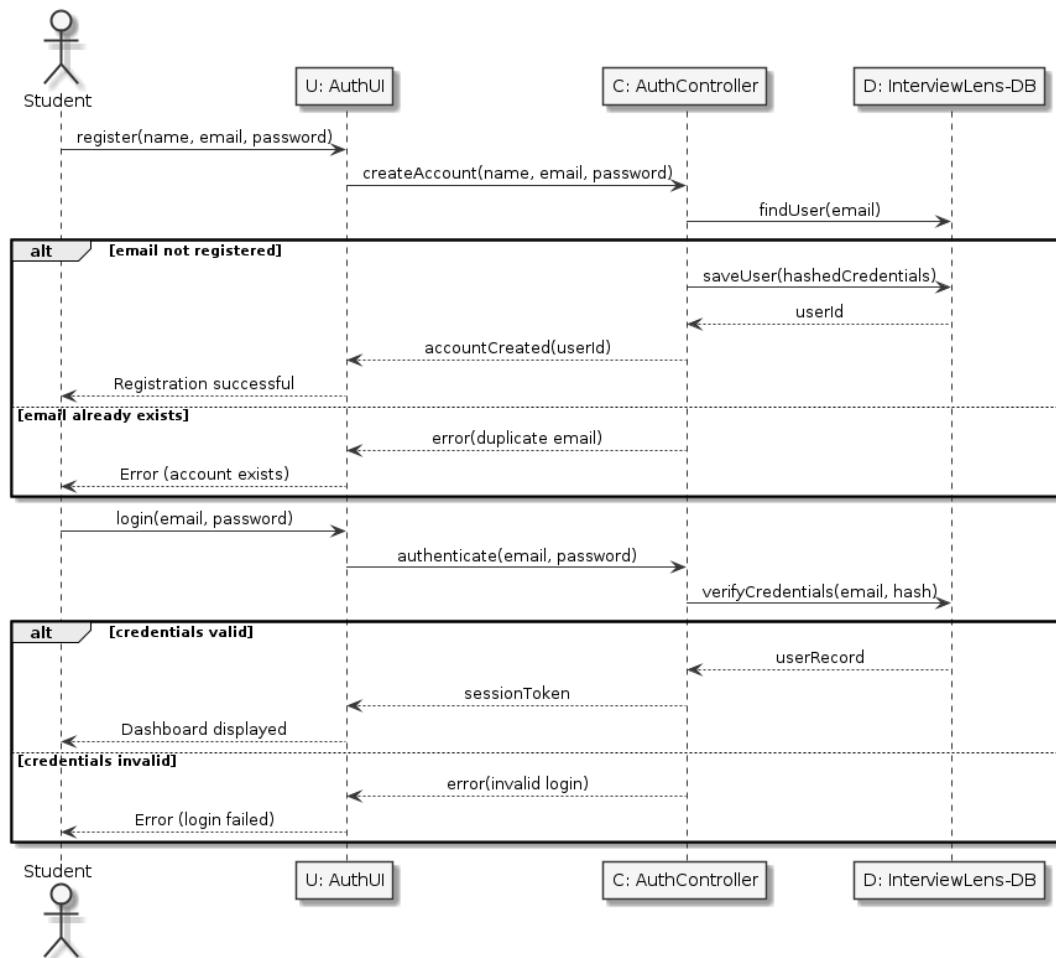


Figure 2. Sequence diagram for Register / Log In

3.6.2 Conduct Mock Interview. This is the representative operation for the whole product. A loop fragment covers the question cycle and an alt fragment handles the adaptive follow up, which is FR-3 in action.

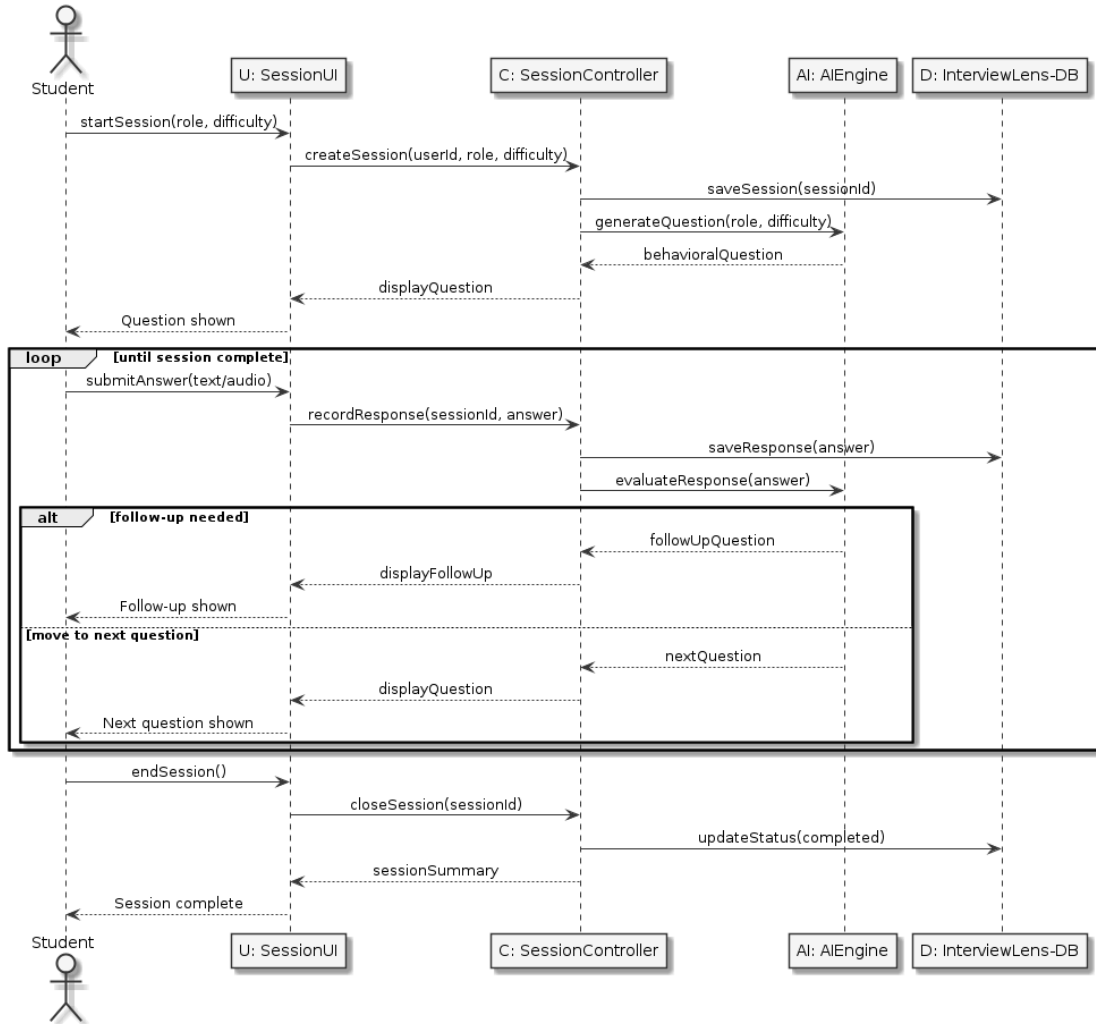


Figure 3. Sequence diagram for Conduct Mock Interview

3.6.3 Receive AI Feedback Report. The controller retrieves the recorded responses, sends them to the AI engine for scoring on STAR structure, clarity and confidence, saves the report, and returns it. An alt fragment handles analysis failure.

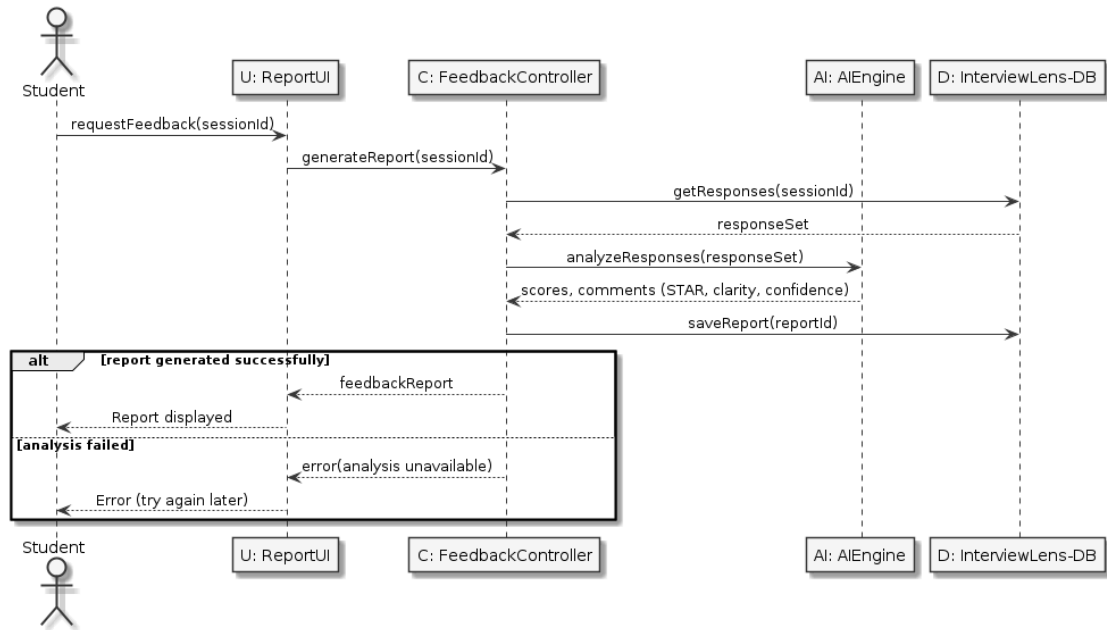


Figure 4. Sequence diagram for Receive AI Feedback Report

3.6.4 View Progress History. The controller loads the student's sessions and computes score trends over time, with an alt fragment for the case where no sessions exist yet.

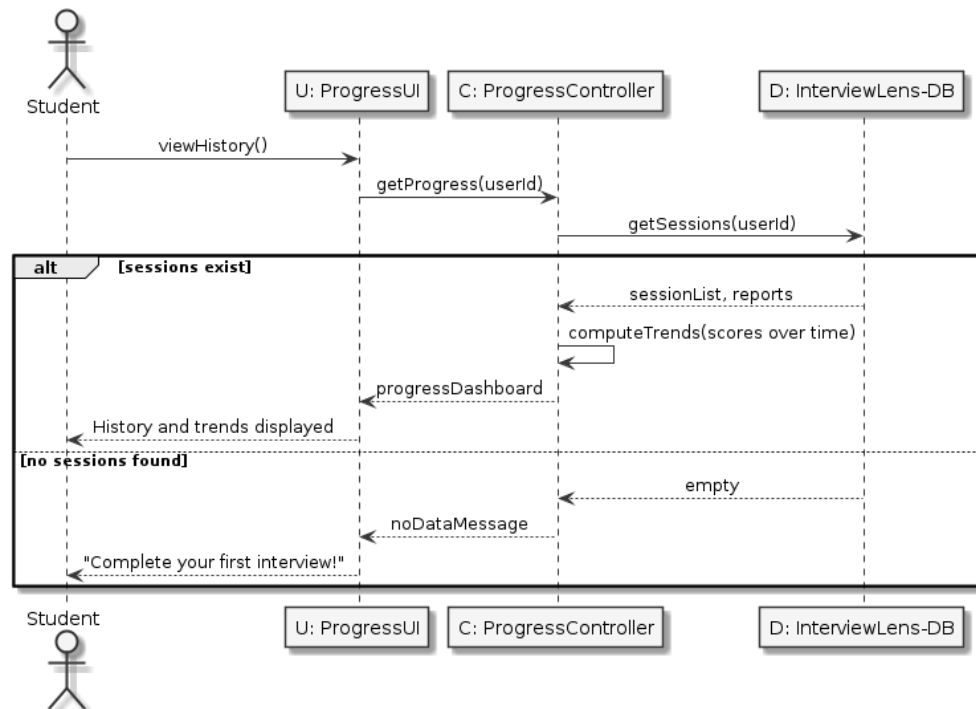


Figure 5. Sequence diagram for View Progress History

3.6.5 Manage Question Bank. An administrator adds a question. The request passes through an authorization check before the question is saved, with an alt fragment for authorization failure.

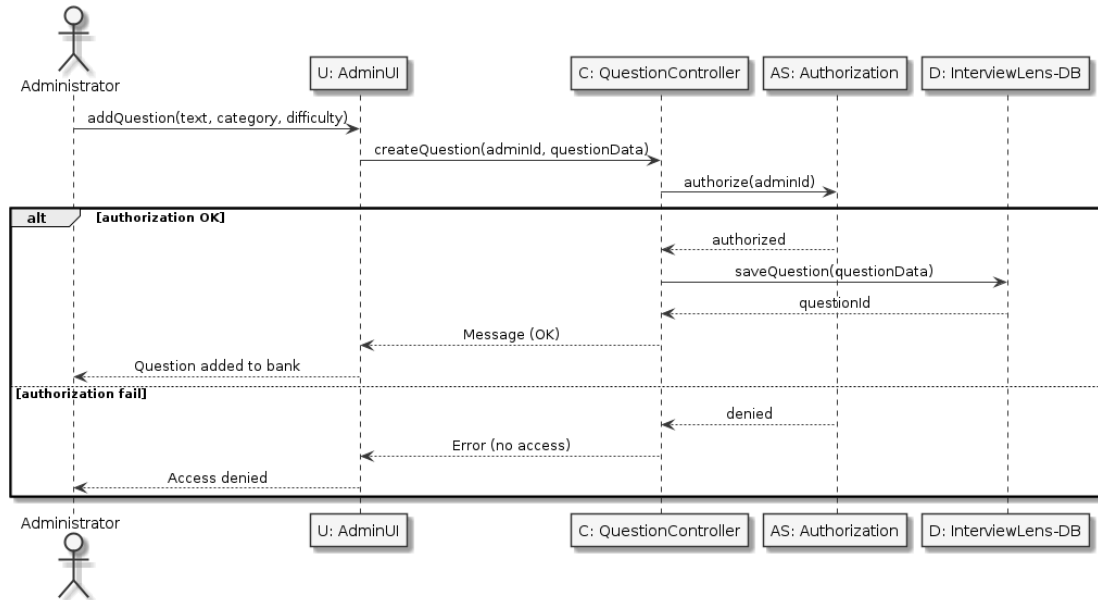


Figure 6. Sequence diagram for Manage Question Bank

3.7 Class Diagram

The class diagram covers all nine classes in the system. Student and Administrator specialize the abstract User class, so authentication and profile behavior are defined once. InterviewSession aggregates the Questions asked in it and composes the Responses recorded in it: the distinction matters because a Response has no meaning outside its session and is deleted with it, whereas Questions are pooled and reused. QuestionBank aggregates that reusable Question pool. AIEngine supports zero or more sessions and generates zero or more FeedbackReports, and each completed session is evaluated by at most one report.

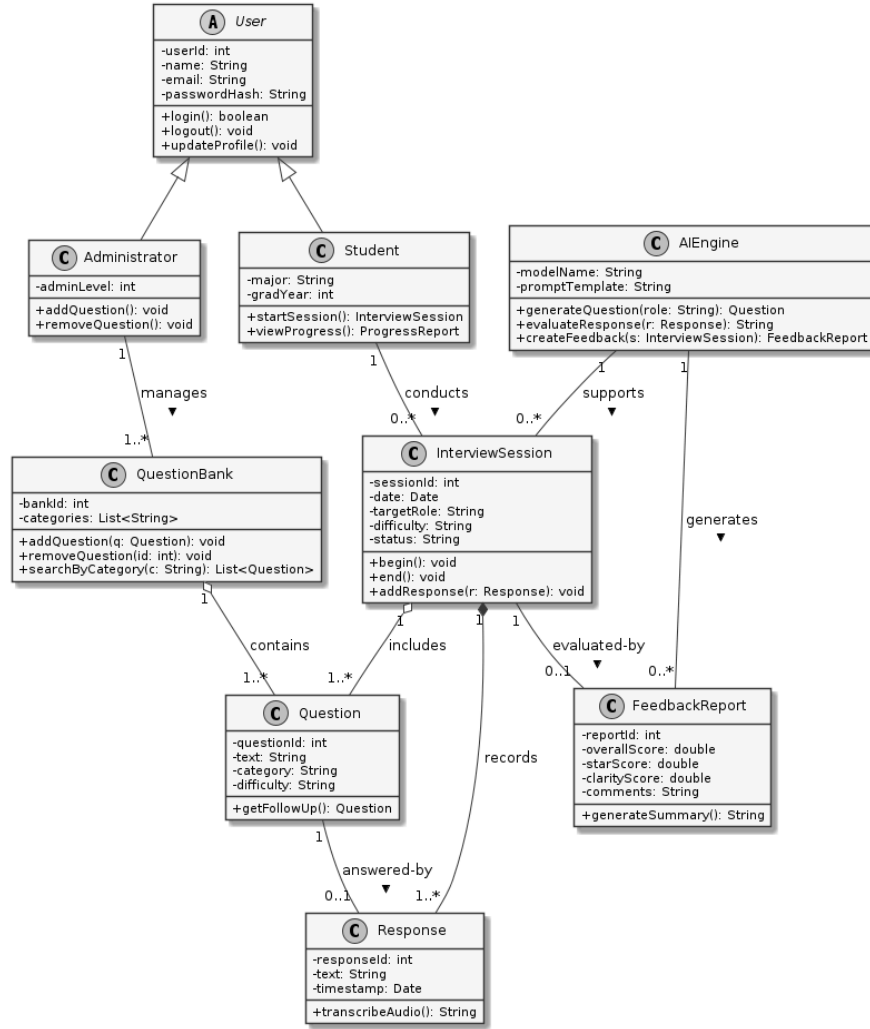


Figure 7. Class diagram for Interview Lens

3.8 Architectural Design

We chose the Model View Controller pattern, described by Sommerville as Figure 6.6 in the architectural design chapter [1]. Interview Lens is an interactive web application in which the same underlying data, meaning sessions, responses and feedback, is presented in three different ways: a live interview screen, a feedback report, and a progress dashboard. Presentation is also the part we expect to change most as we iterate on UI/UX. MVC isolates those presentations in the View, implemented as React components, from the interaction logic in the Controller, which handles HTTP request processing, validation and the session, feedback and auth logic, and from the business logic and data in the Model, which holds the domain classes, the AI engine calls and the PostgreSQL database.

We considered the layered and repository patterns from the same chapter. MVC won because our defining characteristic is one dataset presented several ways. It also maps onto how the team divided the work: frontend work stays in the View, endpoint work in the Controller, and database

plus AI pipeline work in the Model, which lets all five members work in parallel with few merge conflicts.

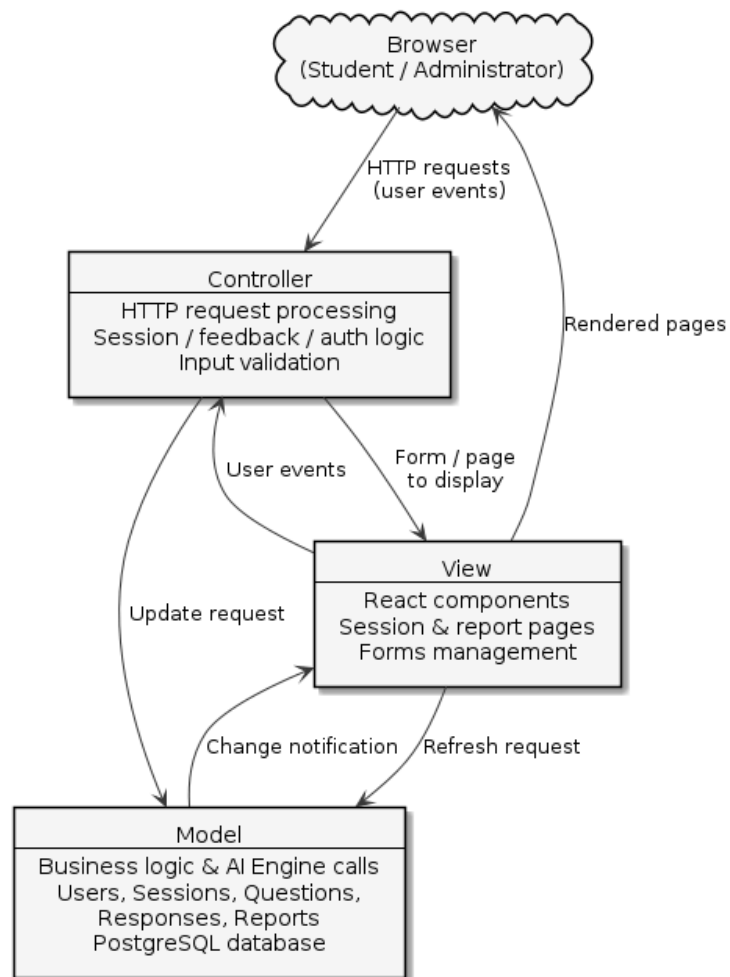


Figure 8. MVC architecture of Interview Lens

4. Project Scheduling, Cost, Effort and Pricing Estimation

Every figure in this section describes building the Interview Lens system itself, not the effort our team spent on coursework.

4.1 Project Scheduling

Start date: Monday, June 1, 2026. End date: Friday, September 4, 2026.

Weekends are not counted. The schedule assumes a Monday to Friday working week, and weekend days are excluded from the working day count entirely rather than held as buffer. Two federal holidays that fall on weekdays inside the window, Juneteenth on Friday June 19 and Independence Day observed on Friday July 3, are also excluded. The window contains 70 weekdays, so removing those two holidays gives 68 working days.

The number of working hours per day is 8 per developer.

Justification. The Function Point analysis in Section 4.2 produces an estimated 2,002 person hours. Divided across five developers at 8 hours per day, that is 50 working days of pure effort. We scheduled 68 working days, which adds roughly a third on top of the raw effort estimate to absorb integration, code review, and AI prompt tuning. Expressed as capacity, five developers across 68 days at 8 hours gives 2,720 person hours against 2,002 hours of estimated effort, leaving 718 hours, or about 26 percent of capacity, as buffer. We deliberately left a buffer that large because the AI pipeline carries the most requirement volatility in the project, which is the same reasoning behind choosing an incremental process model.

The work is organised into four increments, shown below.

Phase	Dates	Duration
Requirements and design	Jun 1 to Jun 19	3 weeks
Increment 1: authentication, database, question bank	Jun 22 to Jul 10	3 weeks
Increment 2: session engine and AI pipeline	Jul 13 to Aug 7	4 weeks
Increment 3: feedback report and progress dashboard	Aug 10 to Aug 21	2 weeks
System test, hardening and user acceptance testing	Aug 24 to Sep 1	1.5 weeks
Deployment and handoff	Sep 2 to Sep 4	3 days

Increment 2 is the longest because the AI pipeline is where the requirements are least stable. Milestones are design freeze at the end of week 3, core release at the end of week 10, feature complete at the end of week 12, and go live at the end of week 14.

4.2 Cost, Effort and Pricing Estimation

Method chosen: Function Point analysis. Of the two permitted techniques we chose Function Point over application composition because Interview Lens is defined by a well bounded set of user recognisable transactions and data stores rather than by assembling prebuilt components. The technique was introduced by Albrecht in 1979 as a way of sizing software from the user's point of view rather than by lines of code [2], and later validated against development effort by Albrecht and Gaffney [3]. We follow the standard five function types and the 14 general system characteristics [4], [5].

Step 1: count the unadjusted function points. Weights are the standard simple, average and complex values. Audio answer submission and the AI feedback report were rated complex, because both involve external service calls and multi field processing. Profile update and question deletion were rated simple.

Function Type	Components Counted	Count	Weighted
External Inputs (EI)	Register, log in, start session, submit text answer, submit audio answer, add question, edit question, delete question	8	30
External Outputs (EO)	Feedback report, progress dashboard, session summary, audit log export	4	24
External Inquiries (EQ)	Session history list, single past report, question bank search, retrieve next question	4	15
Internal Logical Files (ILF)	User, InterviewSession, Question and QuestionBank, Response, FeedbackReport, AuditLog	6	51
External Interface Files (EIF)	LLM provider API, speech to text transcription service	2	14
Unadjusted Function Points		24	134

Step 2: compute the value adjustment factor. Each of the 14 general system characteristics is rated from 0 to 5 and summed to give the total degree of influence, which is then converted using $VAF = 0.65 + (0.01 \times TDI)$ [4]. Our ratings sum to a TDI of 50, giving a VAF of 1.15. The characteristics we rated highest, at 5 out of 5, were data communications, online data entry and complex processing. Complex processing scored 5 because the AI engine performs transcription, rubric scoring and adaptive question generation. Facilitate change scored 4 to reflect the requirement volatility already noted in our process model rationale.

Step 3: derive effort and price.

Step	Value
Unadjusted function points (UFP)	134
Total degree of influence (TDI)	50

Step	Value
Value adjustment factor (VAF = $0.65 + 0.01 \times 50$)	1.15
Adjusted function points ($134 \times 1.15 = 154.1$)	154
Productivity rate assumed	13 person hours per function point
Total effort (154×13)	2,002 person hours

The productivity rate of 13 hours per function point is our own assumption for a small team building a TypeScript web application with third party AI services. We note this is the least certain input in the whole estimate. Albrecht and Gaffney's validation work, and the later critiques of function point analysis, both stress that the conversion from function points to effort is organisation specific and has to be calibrated against a team's own historical data [3], [4]. We have no historical data, so this figure should be treated as an assumption rather than a measurement.

Pricing. Adding the three cost categories detailed in Sections 4.3 to 4.5 gives a total development cost of \$128,243. We apply 15 percent for overhead and contingency and a 20 percent profit margin.

Item	Amount
Total development cost	\$128,243
Overhead and contingency at 15%	\$19,236
Subtotal	\$147,479
Profit margin at 20%	\$29,496
Quoted project price	\$176,975

Dividing the \$128,243 development cost by 154 adjusted function points gives a cost of \$833 per function point, which is the figure we would carry forward to size any future release.

4.3 Estimated Cost of Hardware Products

Interview Lens is browser based and deployable on commodity Linux cloud instances per NFR-6, so most of this category is rented cloud infrastructure priced over a 12 month first year of operation rather than purchased equipment.

Item	Basis	Cost
Developer workstations	5 at \$1,600	\$8,000
Application server instances	2 at \$140 per month	\$3,360
Managed PostgreSQL instance	\$95 per month	\$1,140

Item	Basis	Cost
Object storage for audio recordings	1 TB at \$23 per month	\$276
Load balancer and bandwidth	\$45 per month	\$540
Staging environment and backup storage	\$100 per month	\$1,200
Total hardware		\$14,516

4.4 Estimated Cost of Software Products

The two largest lines here scale with usage rather than with headcount, because every interview session makes LLM and transcription calls. We size them against an assumed 60,000 sessions in the first year.

Item	Basis	Cost
LLM API usage for question generation and scoring	\$0.09 per session, 60,000 sessions	\$5,400
Speech to text transcription	\$0.006 per minute, 180,000 minutes	\$1,080
Monitoring and error tracking	Annual	\$1,428
GitHub Team seats and CI/CD minutes	5 seats, annual	\$660
Figma Professional	2 seats, annual	\$360
IDE licences, TLS certificate and domain	Annual	\$915
Total software		\$9,843

4.5 Estimated Cost of Personnel

Personnel is costed against the 2,002 hours of effort from Section 4.2, allocated across the five roles. Training cost after installation is included as required.

Role	Member	Hours	Rate	Cost
Backend and database engineer	Ayro S. Escobar	420	\$50	\$21,000
Backend and DevOps engineer	Ben Andre Quidlat	420	\$50	\$21,000
AI engineer	Hadeed Sarfaraz	440	\$60	\$26,400
Frontend engineer	Shaun Kenan Moukam	400	\$48	\$19,200
QA engineer	Joshua Asteway	322	\$42	\$13,524
Subtotal, development		2,002		\$101,124
Training after installation	2 admin sessions plus documentation			\$2,760
Total personnel				\$103,884

Personnel accounts for 81 percent of total development cost, which is what we would expect for a software project of this kind. The training line covers two eight hour administrator sessions after installation plus 40 hours of writing user documentation and video tutorials.

5. Test Plan

5.1 Test Plan Description

The unit under test is `computeOverallScore()`, the function in the AI pipeline that turns per answer component scores into the single 0 to 100 session score required by FR-5.

We chose this unit deliberately. The AI engine returns different text for the same input, so the language model call itself cannot be asserted against in a conventional unit test. Our approach is to split the pipeline: the model call is replaced by a stub in CI, and only the deterministic weighting and aggregation layer is unit tested. That layer has to be exactly right, because every trend line on the progress dashboard in FR-6 is computed from its output. If the aggregation is wrong, a student's entire improvement history is wrong.

The approach is unit level white box testing using equivalence partitioning and boundary value analysis, following the testing process in Sommerville [1]. The valid partition is all three component scores within 0 to 10. The invalid partitions are a session with no answers at all, and any component score arriving outside 0 to 10 from a malformed AI response.

Tooling is TypeScript with Jest, chosen because it bundles the test runner, assertion library, mocking utilities and coverage reporting in one dependency [11]. Tests run locally and on every pull request through GitHub Actions, and the pipeline fails below 70 percent line coverage, which is what satisfies NFR-8. Pass criteria are that all defined test cases pass, module line coverage is at or above 70 percent, and no unhandled exception escapes the function for any input in the defined partitions.

5.2 Unit Under Test and Automated Test Execution

The implementation, in `SessionScorer.ts`, weights STAR structure at 0.5, clarity at 0.3 and confidence at 0.2, averages across answers, and scales to 0 to 100. The weighting favours STAR structure because the structured interview literature treats the completeness of a past behaviour answer as the primary thing being assessed [7].

```
export interface AnswerScore {
  star: number;      // 0 - 10
  clarity: number;   // 0 - 10
  confidence: number; // 0 - 10
}

const W = { star: 0.5, clarity: 0.3, confidence: 0.2 };

export function computeOverallScore(scores: AnswerScore[]): number {
  if (scores.length === 0)
    throw new Error('Session has no answers');

  const total = scores.reduce((sum, s) => {
    const parts = [s.star, s.clarity, s.confidence];
    if (parts.some(v => v < 0 || v > 10))
      throw new RangeError('Score out of range 0-10');
    return sum + s.star * W.star
      + s.clarity * W.clarity
      + s.confidence * W.confidence;
  }, 0);
  return total;
}
```

```

    }, 0);

    return Math.round((total / scores.length) * 10);
}

```

Running the suite with `jest --coverage` produced the following. Three cases passed and one failed.

```

PASS   computeOverallScore
  v TC-01 valid 5 answer session
  v TC-02 perfect session -> 100
  v TC-03 empty session throws
  x TC-04 rejects out of range component

  * TC-04 rejects out of range component
    Expected: RangeError
    Received: 96

Tests:   1 failed, 3 passed, 4 total
Coverage: 78.4% lines

```

5.3 Test Cases and Results

ID	Input or condition	Expected result	Actual result	Outcome
TC-01	Valid 5 answer session, every answer scored star 8, clarity 7, confidence 6	73	73	Pass
TC-02	Perfect session, all three components at 10	100	100	Pass
TC-03	Empty session, zero answers submitted	Throws Error	Throws Error	Pass
TC-04	Malformed AI payload with star = 12, above the valid range	Throws RangeError	Returned 96	Fail

TC-01 verification: $(8 \times 0.5) + (7 \times 0.3) + (6 \times 0.2) = 7.3$, and $7.3 \times 10 = 73$.

Defect DEF-01. TC-04 is the case that earned its place in the suite. The first implementation applied the weights without validating the component ranges, so a malformed AI response carrying a star score of 12 silently produced a session score of 96 instead of being rejected. Because the score looks plausible, nothing downstream would have caught it, and the corrupted value would have been written into the student's progress history. We added the range guard shown in Section 5.2, re-ran the suite, and all four cases now pass at 78.4 percent line coverage, above the 70 percent gate in NFR-8. This is the concrete payoff of testing the deterministic layer separately from the model: a test suite that only asserted that the AI returned something would not have found this.

6. Comparison with Similar Designs

We surveyed the interview practice tools that students actually reach for, and checked the current status of each rather than relying on general knowledge. Two findings changed our design argument, and both are reported here.

6.1 Google Interview Warmup, retired April 2026

Google launched Interview Warmup in 2022 through its Grow with Google initiative as a free, browser based tool that asked five expert written questions, transcribed spoken answers using machine learning, and surfaced insights such as job related terms and most used words. Google was explicit that the tool did not grade answers, and that the AI transcribed responses rather than generating the questions [8].

That tool no longer exists. Google retired it in April 2026, and the notice on Google's own skills platform reads only that they were doing some spring cleaning and had decided to retire Interview Warmup [9]. The original URL now redirects to a general interview preparation article. This matters for our project in two ways. It removes the most widely used free option from the market, which strengthens rather than weakens the motivation for a free, campus deployable tool. It also removed the comparator we had been treating as our closest analogue, so the rest of this section compares against tools that are currently live.

6.2 Pramp, now Exponent Practice

Pramp is a peer to peer platform where two candidates take turns interviewing each other. It was acquired by Exponent, and since June 2024 all new sessions are booked on the Exponent Practice platform [10]. The model gives genuine human feedback at no cost, and acting as the interviewer builds judgement in a way solo practice does not.

The limitation is structural rather than technical. Feedback quality depends entirely on which peer you are matched with, sessions have to be scheduled in advance around another person's availability, and the free tier is capped. For a student who wants to run through three attempts at the same question on a Tuesday night, peer matching is the wrong shape of tool.

6.3 interviewing.io

interviewing.io offers anonymous mock interviews with senior engineers from large technology companies, and the platform's own FAQ states that interviews start at \$179 per session [11]. The feedback is high quality because it comes from people who conduct real hiring. The platform has also added an AI interviewer for coding and system design.

The barrier is cost and focus. At \$179 and up per session, repeated weekly behavioral practice is not realistic on a student budget, and the platform's centre of gravity is technical interviewing rather than behavioral.

6.4 Big Interview

Big Interview is licensed to universities and is the closest competitor to what we are building, because it reaches students through their institution rather than direct to consumer. In September 2025 it launched PracticeAI, described as a conversational, voice to voice tool that draws on resumes and job descriptions to generate tailored questions and probing follow ups [12].

This finding forced us to revise a claim. We had been describing adaptive follow up questioning as the feature that distinguished Interview Lens. It is not. PracticeAI already does adaptive follow ups, and does them by voice. Our honest position after checking is narrower and more defensible: adaptive questioning is now table stakes among funded competitors, and our differentiation has to rest on the measurement layer instead.

6.5 Design decisions and where Interview Lens differs

Against currently live tools, three things distinguish our design.

- A fixed, published scoring rubric. Every session is scored on the same three dimensions, STAR structure, clarity and confidence, producing a 0 to 100 score that is comparable across sessions. Competitor feedback is largely narrative, or scores delivery signals such as filler words and eye contact rather than the content structure of the answer.
- Longitudinal measurement. Because the rubric is fixed, scores can be trended. The progress dashboard in FR-6 exists to answer the question a student actually has, which is whether they are getting better, rather than how one answer went.
- Educational records handling. Transcripts and feedback are treated as FERPA aligned educational records, private by default. FERPA governs education records held by institutions that receive Department of Education funding [13], and a tool intended for campus adoption has to be built to sit inside that obligation rather than bolt it on.

The rubric design is grounded in the structured interview literature rather than chosen arbitrarily. Levashina and colleagues, reviewing twenty years of empirical research, report that the superiority of structured over unstructured interviews is one of the most consistent findings in the field, and identify both rating scale development and probing and follow up questioning as core structural components [7]. The U.S. Office of Personnel Management makes the same point in its guidance on structured interviewing, and notes that behavioral description interviews show higher validity where the work is complex [14]. A tool that scores a student against a consistent rubric and probes with follow ups is imitating what the research says an effective structured interview does. That is the argument for our design, and it is a stronger one than any single feature comparison.

Assumption behind the positioning: we assume no currently live tool combines a fixed comparable rubric, longitudinal score tracking, and FERPA aligned handling in a product that is free at the point of use for the student. That combination, rather than any individual feature, is our design decision.

7. Conclusion

Deliverable #2 built on the design established in Deliverable #1 by adding the estimation, scheduling and verification work. Three things changed relative to what we originally planned.

Feedback scope, narrowed from free text to a fixed rubric. We originally described the AI feedback as written improvement suggestions in an open format. We narrowed it to a fixed rubric of STAR structure, clarity and confidence with a numeric score. Free form critique reads better once, but it cannot be compared across sessions, cannot be trended, and cannot be unit tested. The narrowing is what made both FR-6 and the entire test plan possible, and Section 6 shows it is also where our remaining differentiation sits.

Features deferred. University single sign on, covered by assumption A5, and non English interviews, covered by A1, were both deferred beyond the first release. Both represent real demand, but neither is needed to validate the core interview loop.

Our comparison work also corrected two things we had believed. Google Interview Warmup, which we had treated as the closest free analogue, was retired in April 2026. Big Interview's PracticeAI already offers the adaptive follow up questioning we had been claiming as our distinguishing feature. We have revised the claim rather than kept it, and Section 6 states the narrower position we can actually defend.

What did not change is as informative as what did. The incremental process model, the MVC architecture, the class model and all twelve non-functional requirements survived Deliverable #2 without revision. The estimation work did not force a redesign, which we take as evidence that the Deliverable #1 design was sound. The one place the estimate should be read with caution is the productivity rate of 13 hours per function point, which is an assumption we could not calibrate against historical data.

Interview Lens remains true to its original motivation. Students can be technically strong and still lose an offer in the behavioral round, and the goal is to give them an accessible way to practise that round and build the people skills that technical ability alone does not demonstrate.

8. References

Every source below was retrieved and checked directly. Accessed July 26, 2026.

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